Natural Computing SoSe25

Exercise Sheet 3 — June 5th, 2025

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1 Quantum Computing Basics

(a) Which of the following are valid quantum states? For each valid state among the formulas below, give the probabilities of observing 0 and 1 when the system is measured in the standard computational basis.

(i) $|a\rangle = \frac{1}{\sqrt{2}}(|1\rangle + |0\rangle)$

(ii) $|b\rangle = 0.6|0\rangle + 0.4|1\rangle$

(iii) $|c\rangle = \frac{\sqrt{3}}{2}i|0\rangle - \frac{1}{2}|1\rangle$

(b) Write the state $|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{i}{\sqrt{2}}|01\rangle$ as two separate qubits.

2 Simple Quantum Circuits

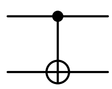
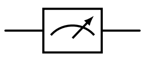
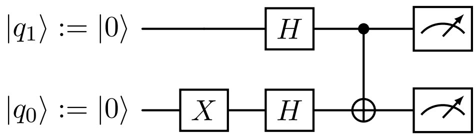
Operator	Gate	Matrix
Identity (I)		$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
Hadamard (H)	H	$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Pauli-X (X)	X	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Controlled-X (CNOT)		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
Measure		

Table 1: Quantum Gate Operators

In the lecture we have introduced our abridged but complete definition of quantum computing (QC). Let's try to apply our formal understanding to some practical examples.[1]

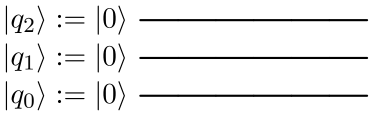
[1] But you may of course still consult `https://quantum.ibm.com/composer`. However, note that the IBM Quantum Composer sorts the qubit wires differently.

(a) Using the quantum operators we have covered in the lecture (cf. Table 1), *calculate* the expected measurement probabilities.
Hint: We can compute the application of a 2-qubit gate G to the qubits q_1, q_0 via $G\,|q_1 q_0\rangle$.

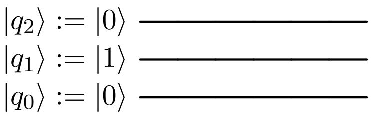


(b) Complete the circuits with the gates given in Table 1 so that the following states would be prepared (given that we measure all states):

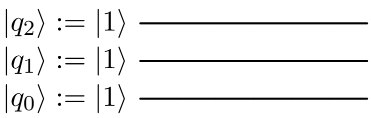
(i) $\frac{1}{\sqrt{2}}|000\rangle + \frac{1}{\sqrt{2}}|011\rangle$



(ii) $|001\rangle$



(iii) $\frac{1}{\sqrt{4}}|000\rangle + \frac{1}{\sqrt{4}}|001\rangle + \frac{1}{\sqrt{4}}|100\rangle + \frac{1}{\sqrt{4}}|101\rangle$



(c) (Bonus) What final state distribution can we expect the following circuit to produce? What are the respective quantum states (i.e., amplitudes) at the end of the four sections?